CHAPTER 8

Optimized Indexes For Fast Retrieval

In the previous chapter you learned how Indexes can be used for filtering operations. Now you're ready to do more with Indexes, going into greater depth on the types of indexes available in PostgreSQL, which fields to cover, and which types are suited best for certain queries. You'll expand on how indexes can be general purpose or have specialized purposes.

You'll create single-column indexes, multicolumn indexes, partial indexes, indexes on expressions, and Covering indexes. As usual, there is a lot of new terminology to learn with Indexing.

Besides speeding up queries for data retrieval from filter operations, indexes are used to enforce `UNIQUE`, `PRIMARY KEY`, and `EXCLUSION` constraints. Indexes store data in sorted order so they can also be used for `ORDER BY` operations.

Take a look at some of the Indexing terminology you'll encounter.

Indexes Terminology



- Index Definition — How the Index was created
- Multicolumn Indexes — Index definitions that cover multiple columns
- Partial Indexes — Index definitions with an expression that reduces the covered rows
- Covering Indexes — Index definitions that cover all columns for a query

You'll work with some less common Index types, but first up is a refresher on the default Index type, which is the most common type.